

Revision O / O-M02 — JLC manufacturability review

Checked 2026-09-15. Layout O is retained with the requested 4 mm bosses (1 mm above the plate). O-M01 remains archived. O-M02 is a detailed quotation package, not confirmation of supplier acceptance or a pressure-rated release.

Conclusion

The faceplate is a plausible conventional flat CNC part. The body is geometrically machinable in principle, but JLC acceptance remains conditional on deep-hole drilling, sealing finishes and the declared POM grade. Internal edge deburring is excluded at the operator's request; only external deburring and internal chip cleaning are required. No supplier upload or order has been made.

JLC requirements applied

Upload each part separately as one STEP solid with its matching-name PDF. The plate ZIP also contains its cut-profile DXF. Select Threads YES for the body and NO for the faceplate. Threads are represented by tapping pilots in STEP. Order-page options must match the drawings. Tell JLC that these two separately quoted parts mate. [JLC ordering guidance](#)

Both parts fit the published 1100 × 600 × 500 mm envelope. The 3 mm faceplate is at the listed minimum thickness for a flat metal part; its long, perforated form still needs support and flatness control. [JLC capabilities](#)

Thread depths are short: 8 mm full form for G 1/4 and 10 mm for M4. The M4 pilot provides run-out space and a conventional drill point. The R1 boss roots are shallow but require a suitable toolpath; confirm JLC can retain the specified radius. Wall reserves exceed its generic plastic guidance, which is a machining rule, not a pressure test. [JLC CNC design guide](#)

Threads and long bores

The body has 24 G 1/4 and 6 M4 threaded features. Confirm availability of ISO 228 G tooling and gauges; standard BSPP is not a permission to substitute ISO 7 Rp. The final thread gauge requirement controls the threaded region. [JLC thread guide](#)

Our geometry gives $410/11.8 = 34.75D$ for a single bore, or $207/11.8 = 17.54D$ for opposed drilling. These are calculated ratios, not JLC published limits. Special drilling and chip control are likely; nominal straight STEP cylinders do not demonstrate achievable alignment. [Deep-hole tooling context](#)

Material selection

Body requirement: Black unfilled POM; POM-C or POM-H acceptable. Declare grade and supply datasheet; material shall conform to its declared grade. JLC lists POM; confirm the supplied grade and black-stock availability. Branded Delrin is not required. [JLC POM page](#)

The faceplate is dry 304 stainless, a listed CNC material. It is not a wetted rear cover. [JLC SUS304](#)

Fit, edge treatment and supplier questions

Boss clearance

Retain $\varnothing 32 +0.20/0$ plate windows and $\varnothing 28 \pm 0.10$ bosses. Nominal radial clearance is 2 mm. The R1 root grows to $\varnothing 30$ nominal at the POM mounting plane. At worst specified sizes ($\varnothing 28.1$ and $R1.1$), its envelope is $\varnothing 30.3$: 0.85 mm radial reserve before positional error. Independent ± 0.10 X/Z coordinates on both parts consume up to 0.283 mm, leaving about 0.567 mm with aligned assembly frames. Screw-clearance registration, thermal growth and deformation are additional; perform an assembly trial. A $\varnothing 29$ window would interfere with the nominal root.

Boss height is 4.0 ± 0.10 mm above the POM mounting plane. With a 3.0 ± 0.10 mm plate, nominal projection is 1 mm (0.8–1.2 mm size stack before flatness and assembly effects). This clears the plate for fittings whose sealing shoulder seats on the POM; verify unusually recessed shoulders and tool access. Boss lips receive $C0.5 \times 45^\circ$. Their flat sealing annulus remains about $\varnothing 13.8$ – $\varnothing 27$. The existing mouth chamfer is retained; do not enlarge it casually into the O-ring footprint. POM can be burred, dented or gouged at thin edges; the edge break improves handling without treating the material as inherently brittle. Sealing lands require careful finishing, not general rounding.

Faceplate countersinks

Six $\varnothing 4.5$ through holes receive $\varnothing 8 \times 90^\circ$ countersinks on the front only. Nominal depth is 1.75 mm in 3 mm stock, leaving 1.25 mm cylindrical land. The specified DIN 7991 reference has $\varnothing 7.96$ maximum head; DIN 965 Z Pozi has $\varnothing 7.5$ nominal head. Inspect using the specified screw family rather than relying only on a generic “M4 countersunk” label. [Hex head reference](#); [Pozi reference](#)

Sheet-metal alternative

Laser profiling plus secondary countersinking is possible if the drawing tolerances are met. JLC sheet-metal guidance lists minimum hole diameter $\max(1 \text{ mm, half stock thickness})$ and 1 mm between holes; this pattern exceeds those values. Its default tolerances and submission rules differ from CNC. Request the secondary countersinks and specified flatness explicitly. No bending is required. [JLC sheet-metal guide](#)

Confirm with JLC before manufacture

- Can you make two $\varnothing 11.8 +0.08/0$ bores through 410 mm in the specified black POM, including the 0.30 mm alignment / meeting-step requirements?
- What drilling and inspection method will be used? Internal edge deburring is not required; passages must be cleaned/flushed free of loose machining chips.
- Can you supply the specified resin grade and machine/gauge G 1/4 ISO 228-1 and M4-6H directly in POM?
- Can you achieve the indicated seal-face finish, local flatness, perpendicularity and plate flatness without rounding the sealing lands?
- Please review the two-part fit and report any proposed geometry, tolerance or material changes before proceeding.

Remaining design limits

The approved rack pattern retains 1.9 mm nominal metal at the outermost slot edges; washer overhang there remains a builder-hardware issue. The previous 2 mm elbow/nut gap uses illustrative envelopes. Actual hardware, temperature, clamp preload, creep and assembled leak/pressure behaviour have not been qualified. These findings do not prevent requesting a manual manufacturing review.